CV - Giotto Frean

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Summary

PhD student at the University of Manchester working on decision-making for autonomous sensing systems, particularly under energy constraints. Work experience in developing acoustic classification models for ecological conservation, agent-based modelling, and low-latency inference.

Education

- **BSc (Hons) Computer Science, First Class**
  Victoria University of Wellington, Jan 2020 – Oct 2020
  Completed a research project on Coarse-To-Fine machine learning strategies. Explored the limitations of ensembles, fractal networks, and node splitting methods.
  GPA equivalent: 3.89 / 4.00.

- **BSc Computer Science & Politics**
  Victoria University of Wellington, Jan 2017 – Oct 2019
  Double major, focused on Computer Science.

Research and Technical Projects

- **My own research**

  - *Energy Efficient Marine Inference:* Explored machine learning and decision-making algorithms for an autonomous underwater vehicle (AUV), focusing on energy efficiency, low-bandwidth communication, passive acoustic sensing with hydrophone arrays, and species recognition. Github.

  - *Multi-agent Systems:* Implemented key algorithms and developed notebooks covering distributed constraint satisfaction, distributed optimization, non-cooperative game theory, sequential games, congestion games, and communication. Following the book *'Multiagent Systems'* by Yoav Shoham and Kevin Leyton-Brown. Github.

  - *Algorithms for Decision Making:* Implemented key algorithms and developed notebooks covering PGMs, MDPs, EM, sampling, Bayesian decision theory, and Value of Information. Following the book *'Algorithms for Decision Making'* by Mykel J. Kochenderfer, Tim A. Wheeler, and Kyle H. Wray. Github.

- **Viseme learning**
  Designed a computer vision system to extract visemes from video data, combining phoneme recognition with neural facial representation models. Made short animations with this tool. Github.

- **Quitter**
  Built and deployed *Quitter* (https://quitter-web.onrender.com/about), a social media prototype where content is ranked via quadratic voting across repeated rounds. Users allocate positive or negative votes using a fixed credit budget, aligning outcomes with true preference rather than raw engagement. Built with Dash and released open-source on Github.

Publications

- **Optimising policy to achieve environmental goals for agriculture**
  SSRN, May 2024
  Developed a hectare-level multi-agent model of New Zealand agricultural land use. Analysed the impact of emission pricing and capability-building policies on emissions, water quality, and profitability. Published here.

- **Mitigating Domain Shift in Bird Call Recognition**
  Forum Acusticum, Sept 2026
  Examined the impact of domain shift in bird call recognition, and designed some novel normalization strategies. Published here.

Technical Skills

- **Programming**
  Python (TensorFlow, PyTorch, SciPy, NumPy, Pandas, Matplotlib, Scikit-learn), Julia, SQL, Java, C#.
- **Machine Learning & AI**
  Bayesian networks, reinforcement learning, linear models, HMMs, CNNs, transformers, Gaussian Processes, kernel regressions, MCMC sampling, value iteration.
- **Multi-agent Systems**
  Distributed optimization and constraint satisfaction, non-cooperative game theory, POMDPs, Nash equilibria, repeated games.
- **Mathematics**
  Linear algebra, graph theory, statistics, discrete mathematics, probability.

Work experience

- **Programmer**
  Victoria University of Wellington, Nov 2024 – Present
  Developing a pipeline for bird species classification based on their calls. Using Audio Spectrogram Transformers, with a focus on the domain shift problem across different recording environments.
- **Quantitative Associate**
  Scarlatti, May 2021 – Apr 2024
  Conducted applied research with economists on the financial and wellbeing impacts of government agricultural policy, using geographical datasets and agent-based simulations. Developed a preference-elicitation model to quantify hard-to-price outcomes (e.g., sense of community), and worked with wellbeing experts to develop a latent-variable framework linking observed behaviours, like multitasking, to health and productivity measures, such as stress.
- **Research Assistant**
  Victoria University of Wellington, Feb 2021 – May 2021
  Conducted signal processing and model fitting on noisy astronomical image data from the Chandra X-Ray Observatory. Background removal, spectral analysis, curve fitting.
- **Data Analyst**
  Sail GP, Nov 2019 – Feb 2020
  Trained Gaussian Process models to predict the physical forces acting on a racing yacht from state variables such as velocity and foil height. This is used by the motor control system to make real-time low-latency decisions.

Research interests

Ecological monitoring with AUVs, population inference, machine learning for conservation. Also multi-agent systems and robotics.

Awards

- **Dean’s List, Victoria University of Wellington**
- **Royal Society Prize, NIWA Wellington Science Fair (2013)**

References

Dr JP Lewis: *Principal Research Scientist* at Electronic Arts, noisebrain@gmail.com
Dr Tulasi Parashar, *Senior Lecturer* at Victoria University of Wellington, tulasi.parashar@vuw.ac.nz